

SWEETY AKTER

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PROFILE

Biotechnology student at BRAC University with a strong focus on bioinformatics, machine learning, and multi-omics integration. Research experience spans computational biology, antimicrobial resistance, and cancer genomics, with work published in IEEE and Springer proceedings. Proficient in Python and R. Currently preparing to transition into quantitative finance, with a long-term goal of pursuing algorithmic trading and financial risk modelling.

EDUCATION

BSc in Biotechnology & Genetic Engineering, BRAC University, Dhaka 2023 – 2027 (Expected)
CGPA: 3.47/4.0 (UK 2:1 equivalent), reflecting a strong upward academic trajectory.

Quantitative Highlights:

- Fundamentals of Mathematics (MAT101) Grade: A+ (4.00)
- Introduction to Statistics (STA101) Grade: A (4.00)
- Biostatistics (BTE317) Grade: B (3.00)
- Introduction to Computer Science (CSE101) Grade: B (3.00)

Additional Quantitative Coursework (In Progress):

- Calculus and Analytical Geometry (MAT211)
- Linear Algebra and Differential Equations (MAT212)
- Probability and Statistics (STA201)

SELECTED PROGRAMMES

GCI World 2026 – AI & Data Science Programme, The University of Tokyo (Matsuo-Iwasawa Laboratory) Expected 2026
• Competitive 3-month online programme in AI and Data Science, selected from 25,000+ applicants across 100+ countries
• Curriculum covers Python programming, machine learning, and business applications
• Programme partnered with the Japan International Cooperation Agency (JICA)

TECHNICAL SKILLS

Programming: Python, R, SQL, Git/GitHub

Quantitative Methods: Bayesian Modelling (MCMC), Statistical Inference, Regression Analysis, Hypothesis Testing, Probability Theory, Calculus, Linear Algebra

Machine Learning: Deep Learning, Graph Neural Networks, Predictive Modelling, Feature Engineering, Dimensionality Reduction (PCA, t-SNE)

Data Visualisation: Matplotlib, Seaborn, ggplot2

Computational Biology: BLAST, GATK, PyRx, Discovery Studio, NCBI, PDB

Tools: Microsoft Excel (Advanced), AWS (Basics), PowerPoint

CONFERENCE PUBLICATIONS & PRESENTATIONS

IEEE Conference Paper (Accepted & Presented):

- Bayesian Beta-Mixture Model for Clonal Evolution Inference from DNA Methylation Data in Haematological Cancers — 2026 IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON 2026), 4–5 September 2026, IUBAT, Dhaka; submitted for inclusion in IEEE Xplore. Used Bayesian statistical modelling (Markov Chain Monte Carlo) and hierarchical Bayesian models to perform probabilistic inference on cancer progression dynamics from high-dimensional DNA methylation data, demonstrating advanced quantitative and computational skills directly relevant to financial modelling.

Springer CCIS Publication (SEMIT 2026 Conference Papers):

- Machine Learning-Based Patient Stratification Using Multi-Omics Data in EGFR-Mutant Lung Adenocarcinoma: A Systematic Review and Framework Proposal — 6th International Conference on Science, Engineering Management, and Information Technology (SEMIT 2026), 16–18 September 2026; accepted for publication in Springer CCIS (Paper ID: 1122-SEMIT 2026 (R2)). Implemented a machine learning framework integrating multi-omics data with feature selection and dimensionality reduction techniques (PCA, t-SNE). DOI: 10.1007/978-3-032-39711-9_23
- Independent Validation and Clinical Risk Stratification of a Pre-trained k-mer-Based Machine Learning Model for Antimicrobial Resistance Prediction Using 18,916 Bacterial Isolates — SEMIT 2026, 16–18 September 2026; accepted for publication in Springer CCIS (Paper ID: 1123-SEMIT 2026 (R1)). Validated ML model performance on large-scale bacterial genomic data and employed statistical methods for clinical risk stratification. DOI: 10.1007/978-3-032-39711-9_24

ADDITIONAL RESEARCH WORKS

Peer-Reviewed Journal Articles:

- Aplasia Cutis Congenita with Fetus Papyraceus (Frieden Classification Type V) — European Journal of Pediatrics, 2026. Systematic review with proposed revised diagnostic criteria; involved data extraction, analysis, and manuscript preparation.
- Integrating AI-Driven Diagnostics in Arrhythmia Care — Cureus, 2026. Narrative review on AI applications in cardiology; contributed to AI/ML methodology sections.
- Fetal Multicystic Dysplastic Kidney and Its Effect on Post-natal Quality of Life — Cureus, 2026. Literature review on pediatric nephrology.

Preprints:

- Deep Learning–Driven De Novo Design of Antimicrobial Peptides Active in Human Serum — DOI: 10.17605/OSF.IO/8ZE2Q
- Precision Targeting of the TP53 R175H Mutation: CRISPR-Cas9 Guide RNA Design — DOI: 10.17605/OSF.IO/58TPC

GITHUB PROJECTS

- Drug-Repositioning-Hub (github.com/sweet-damn/Drug-Repositioning-Hub) — Leveraging AI and the NIH Drug Repositioning Hub to identify new uses for existing drugs
- AMR-Risk-Stratification (github.com/sweet-damn/AMR-Risk-Stratification) — Machine learning framework for antimicrobial resistance risk stratification using multi-omics data from 18,916 bacterial isolates
- computational-biology--streamlit-app (github.com/sweet-damn/computational-biology--streamlit-app) — Integrated platform for molecular simulations, genomic analysis, and cheminformatics
- multiomics-gnn-thesis (github.com/sweet-damn/multiomics-gnn-thesis) — Graph Neural Network for multi-omics disease progression prediction

WORK EXPERIENCE

Student Researcher & Campus Coordinator, BRAC University / Dawn of Bioinformatics Ltd.	2023 – Present
• Published 8 research outputs, including an IEEE conference paper, Springer CCIS papers, and peer-reviewed journal articles • Utilised Bayesian statistical modelling, machine learning, and Graph Neural Networks to analyse 5+ genomic and multi-omics datasets, achieving 92% prediction accuracy • Designed and delivered 15+ workshops on molecular docking, drug design, and computational data analysis, training 100+ students in Python and ML applications • Built 4+ GitHub repositories demonstrating practical coding and data analysis skills • Collaborated with 8+ national and international research partners across 4 interdisciplinary projects	
Peer Reviewer, Briefings in Bioinformatics (Oxford University Press)	July 2026 – Present
• Review 15+ manuscripts annually on computational biology, bioinformatics, and machine learning applications • Provide critical feedback on methodology, statistical analysis, and reproducibility for 15+ manuscripts annually	
Assistant Manager, Youthbees, Dhaka	Dec 2023 – May 2024
• Managed a team of 5 to coordinate 12+ events, improving operational efficiency by 25%	
Educational Counsellor, College Mastermind (Online, USA)	May 2023 – Oct 2023
• Guided 50+ students through biotechnology course admissions and academic planning	

CERTIFICATIONS

- Bioinformatics Course – Plasma Cell Research (2024)
- CRISPR-Cas9 Idea Finalist – BRAC University (2024)
- Duke of Edinburgh Award – Bronze & Silver (2023–2024)
- Leadership Programs – Harvard, UNICEF, HP LIFE (2023–2024)

MEMBERSHIPS

- Director, BRAC University Research for Development Club (2024 – Present)
- Associate Member, UNYSAB Bangladesh (2024 – Present)
- Student Researcher, BRAC University (2023 – Present)

LANGUAGES

Bengali (Native) | English (Fluent) | German (Basic)

REFERENCES

Available upon request.